ChronoFork⏳🔀:

Immersive Learner–Multi-Agent Co-Roleplay for Divergent Historical Experiences

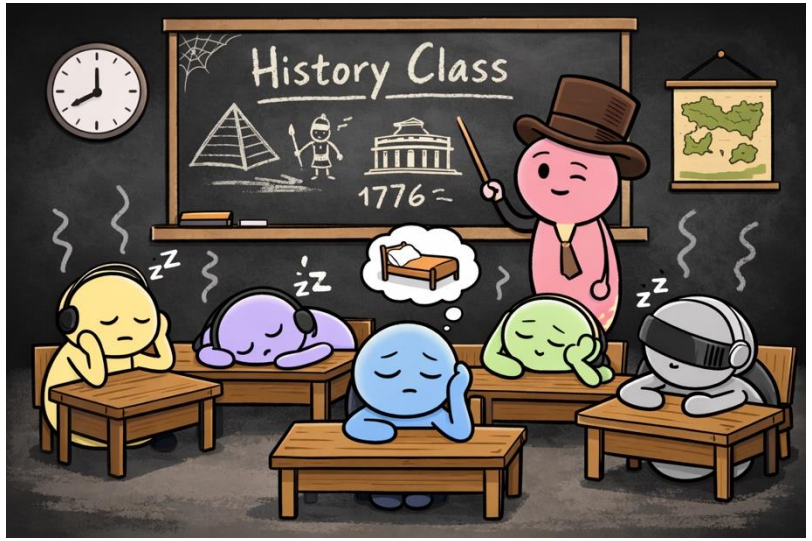
Ziyi Zhang
IS492 - Spring 2026

https://github.com/IS492-SP26/team-project-chronofork-history-rewrite-simulation

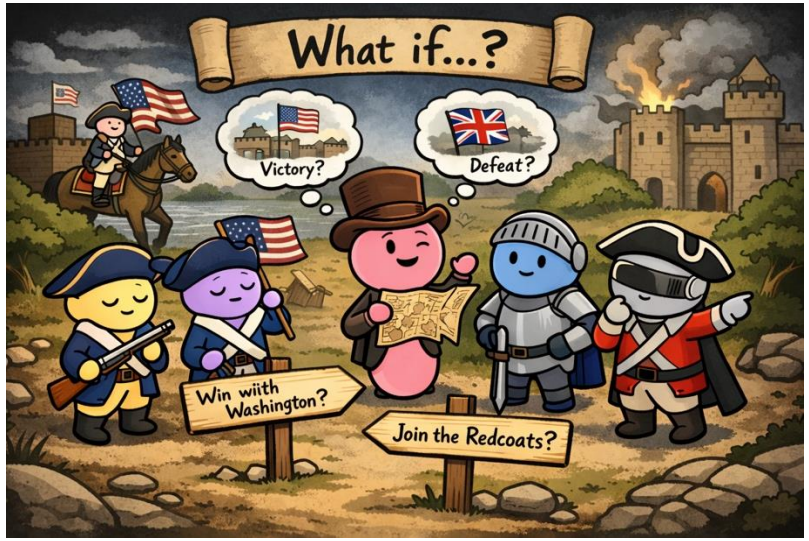
ILLINOIS

1. Problem & Motivation: History Education

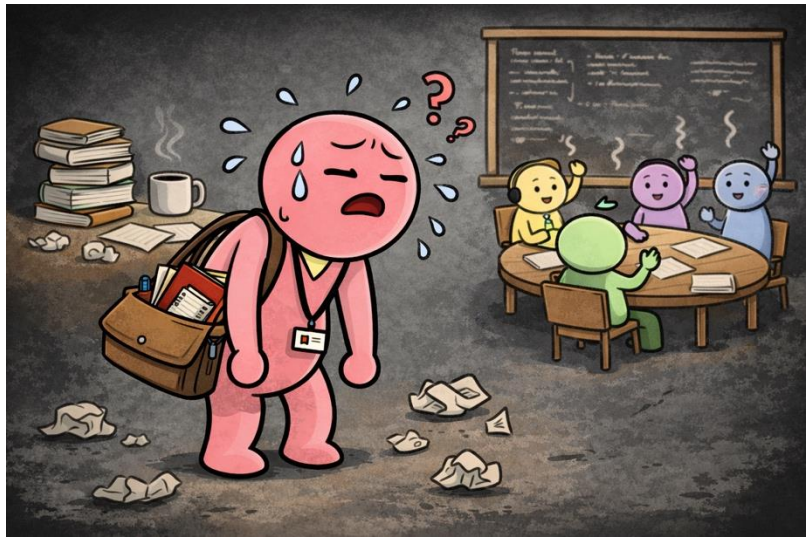
- From “sleepy history class” to historical thinking + counterfactuals (Challenging)



History ≠ memorization
Historical Thinking



Counterfactual helps historical thinking

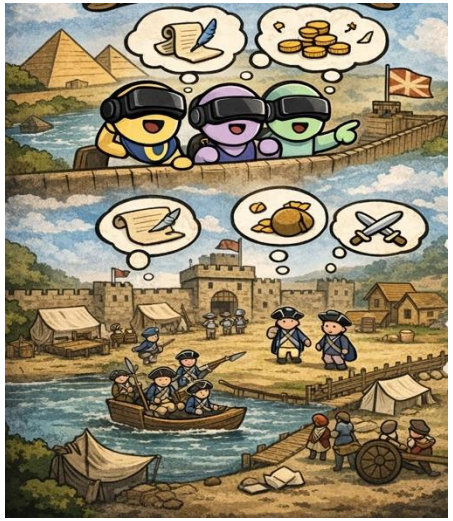


mostly teacher-led
→ hard to scale

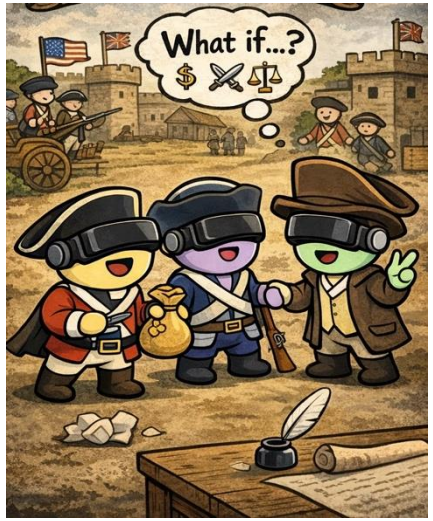
2. Target Users & Core Tasks

Primary users: history learners (students / enthusiasts)

- Stage 1: Observe canonical history (context first)
- Stage 2: Intervene as a figure → diverge history
- Stage 3: Reflect (report + guidance)



→



→

3.1 Related Work: LLM Roleplay Tools

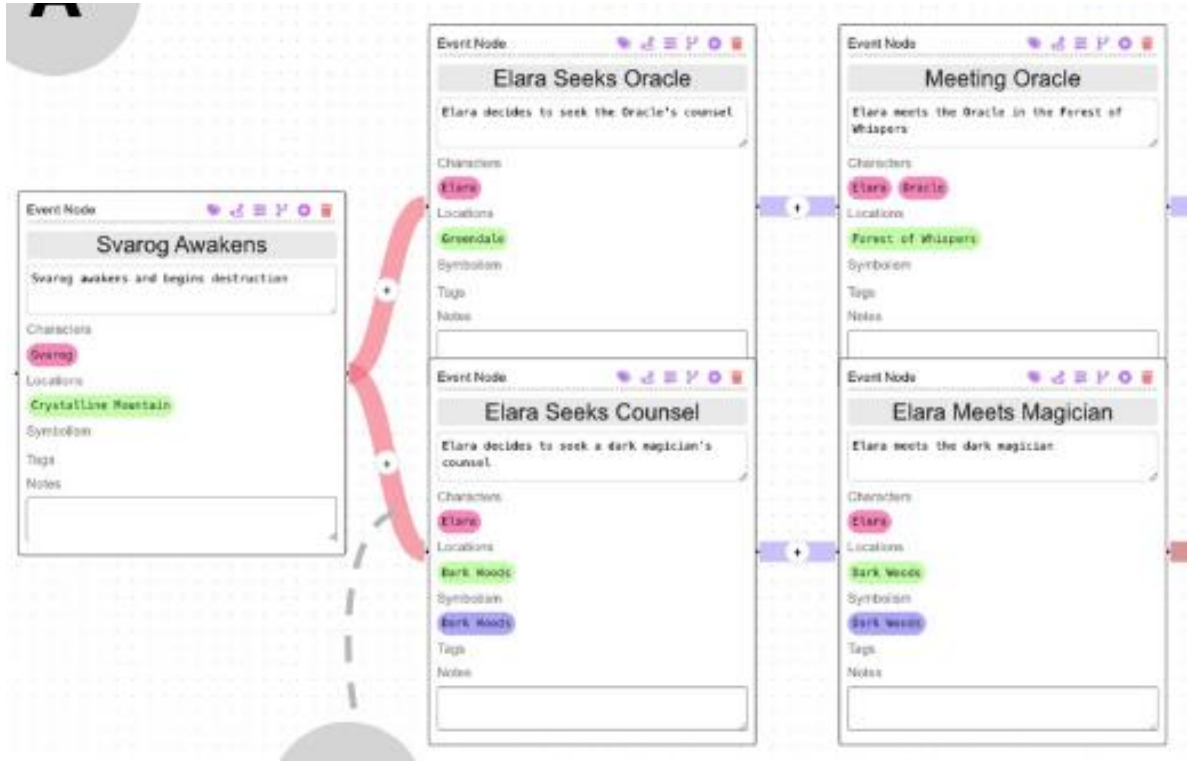
LLM Roleplay Tools for Education

- Insights:
  - LLMs drive **credible characters** ✅
  - Interacting increase **interest & recall** ✅
- Gaps:
  - Most are **passive** / Q&A-driven ❌
  - Learners still **not "inside"** a story world ❌

3.2 Related Work: Narrative Generation

Narrative Generation

- Insights:
  - LLM story gen is **good & fast** ✅
- Gaps:
  - **Offline** + human-out-of-loop ❌
  - **Not dynamic** co-roleplay ❌



3.3 Related Work: VR for education

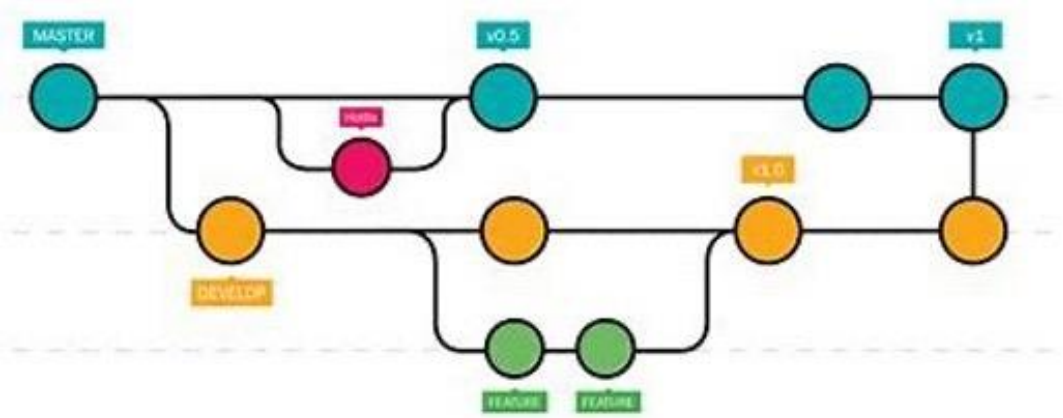
VR for education

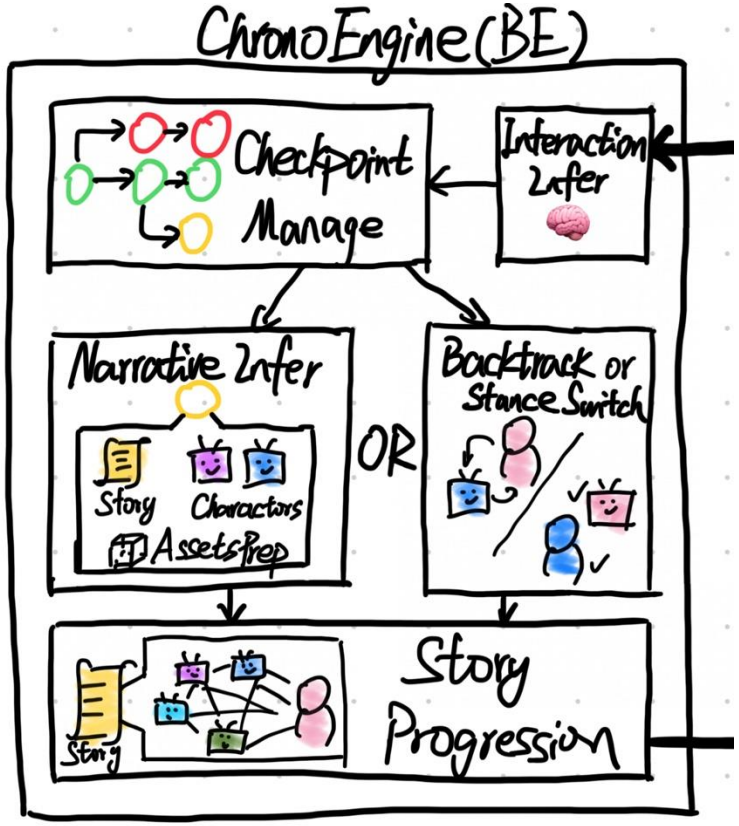
- Insights:
  - VR boosts **curiosity + empathy** ✅
- Gaps:
  - Scenes are curated hand-built ❌
  - Time & cost ❌
- Opportunity:
  - Dynamic GenAI 3D asset generation

4.1 Initial Concepts: ChronoEngine

ChronoEngine : Backend for narrative and roleplay

- Decision checkpoints = Git branches
- Agents push learner into story

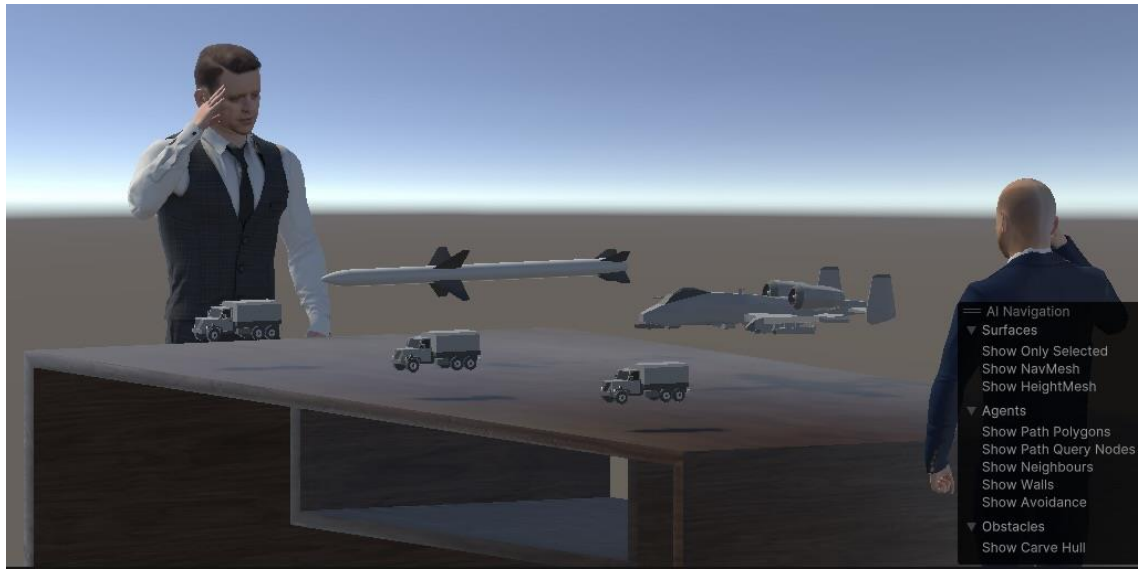


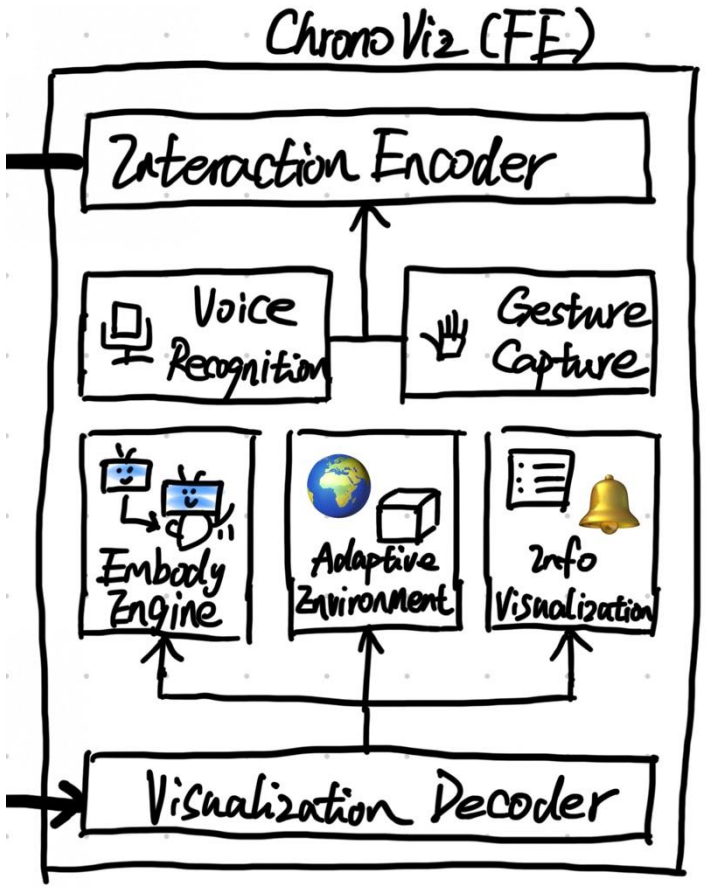


4.2 Initial Concepts: ChronoViz

ChronoViz 🎬: Frontend for Immersive Experience

- Adaptive 3D scene + embodied agents
- Natural interaction, low load visualizations





5. Next Steps, Milestones, Risks

CP2: validate ChronoEngine via prompting

- Can it generate plausible consequences for divergence?

Risk: hallucinations / absurd outcomes

- Mitigation: RAG + plausibility checks

Thank You

Q&A

ILLINOIS